



Lab Sheet 1

Data Structures – I Lab (BTCS209P)

B.Tech.(AIML,CyberSecurity,andDataScience)

III Semester (Session 2026–2027)

Objective

To learn the fundamentals of C++ programming and implement basic programs that forms the foundation of Data Structures.

Prerequisites

Basic knowledge of C++ programming.

A C++ compiler (Code::Blocks / Visual Studio Code / Dev-C++).

Familiarity with compiling and executing C++ programs.

Task

Open your C++ IDE and perform the following tasks.

Task 1: Basic Input and Output

a) Write a C++ program to display:

Welcome to Data Structures Lab

Course Code : BTCS209P

b) Write a C++ program to input two integers and display Sum, Difference, Product, Quotient and Remainder.

Task 2: Decision Making Statements

a) Check whether a number is Even or Odd.

b) Find the largest among three numbers.

c) Check whether a year is a Leap Year.

Task 3: Looping Statements

a) Print numbers from 1 to 100 using for loop.

b) Display multiplication table of a given number.

c) Find factorial using while loop.

d) Generate Fibonacci series up to N terms.



Task 4: Functions

- a) Find square using a user-defined function.
- b) Find maximum of two numbers using a function.
- c) Check Prime number using a function.

Task 5: Arrays

- a) Read 10 integers into an array and display them.
- b) Find largest and smallest elements.
- c) Calculate sum and average.
- d) Search an element using Linear Search.

Instructions

Write each task as a separate C++ program.

Use meaningful variable names and proper indentation.

Include appropriate comments.

Compile and execute each program successfully.

Save each program with an appropriate filename.

Submission

Submit the folder along with the screenshots to the course instructor.

Take screenshots of source code and output for each task.

Save all C++ source files in a single folder.

LABSHEET --1

Task

Open your C++ IDE and perform the following tasks.

Task 1: Basic Input and Output

a) Write a C++ program to display:

Welcome to Data Structures Lab Course

Code : BTCS209P

solution:

```
#include <iostream>
using namespace std;
int main(){
    cout<<"Welcome To Data Structure Lab"<<endl;
    cout<<"Course Code: BTCS209P";
    return 0;
}
```

b) Write a C++ program to input two integers and display Sum, Difference, Product, Quotient and Remainder.

solution: #include <iostream>

```
using namespace std;
int main(){
    int a,b;
    cout<<"Enter first number :";
    cin>>a;
    cout<<"Enter first number :";
    cin>>b;
    cout<<"Sum of two number : "<<a+b<<endl;
    //consider a is greater than b
    cout<<"Difference of two number : "<<a-b<<endl;
    cout<<"Product of two number : "<<a*b<<endl;
    cout<<"Quotient of two number : "<<a/b<<endl;
    cout<<"Remainder of two number : "<<a%b<<endl;
    return 0 ;}
```

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Task 2: Decision Making Statements

a) Check whether a number is Even or Odd. solution:

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout<<"Enter the number: ";
    cin>>num;
    if (num%2==0){
        cout<<num<<" " "<<"is an even number";}
    else {
        cout<<num<<" " "<<"is an odd number"; }
    return 0;
}
```

b) Find the largest among three numbers.

solution:

```
#include <iostream>
using namespace std;
int main() {
    int a,b,c;
    cout<<"Enter three numbers: ";
    cin>>a>>b>>c;
    if ((a>b)&a>c) {
        } cout<<a<<" " "<<"is greater";
    else if((b>a)&(b>c)) {
        cout<<b<<" " "<<"is greater";}
    else {
        cout<<c<<" " "<<"is greater"; }
    return 0;
}
```

c) Check whether a year is a Leap Year. solution:

```
#include <iostream>
using namespace std;
int main() {
    int year;
    cout<<"Enter year: ";
    cin>>year;
    if ((year%4==0)& (year%100!=0) |(year%400==0)) {

        cout<<year<<" "<<"is leap year";}
    else {

        cout<<year<<" "<<"is not a leap year";
    }
    return 0;
}
```

Task 3: Looping Statements

a) Print numbers from 1 to 100 using for loop.

solution:

```
#include <iostream>
using namespace std;
int main() {
    int i;
    for(i=1;i<=100;i++){
        cout<<i<<" ";
    }
    return 0;
}
```

b) Display multiplication table of a given number.

solution:

```
#include <iostream>
using namespace std;
int main() {
    int i,n;
    cout<<"Enter the number:";
    cin>>n;
    for(i=1;i<=10;i++){
        cout<<n<<"x"<<i<<"="<<n*i<<endl;
    }
    return 0;
}
```

c) Find factorial using while loop.

solution:

```
#include <iostream>
using namespace std;
int main() {
    int i,fact=1 ,n;
    cout<<"Enter the number:";
    cin>>n;
    while(n>0){
        fact=fact*n;
        n--;
    }

    cout<<"Factorial of"<<" "<<n<<"="<<fact<<endl;

    return 0;
}
```

d) Generate Fibonacci series up to N terms.

solution

```
#include <iostream>
using namespace std;
int main() {
    int n,a=0,b=1,c,i=1;
    cout<<"Enter the number:";
    cin>>n;
    while(i<=n){
        cout<<a<<" ";
        c=a+b;
        a=b;
        b=c;
        i++;
    }
    return 0;
}
```

Task 4: Functions

a)Find squareusing a user-defined function.

solution:

```
#include <iostream>
using namespace std;
int square(int num)
{
    return num * num;
}
int main(){
    int num;
    cout << "Enter a number: ";
    cin >> num;
    cout << "Square = " << square(num) << endl;
    return 0;
}
```

b) Find maximum of two numbers using a function.

solution

```
include <iostream>
using namespace std;
int maximum(int a, int b){
    if (a > b)
        return a;
    else
        return b;
}
int main()
{

    int num1, num2;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;
    cout << "Maximum = " << maximum(num1, num2) << endl;
    return 0;
}
```

) Check Prime number using a function.

solution :

```
#include <iostream>
using namespace std;
bool isPrime(int number)
{
    if (number <= 1)
        return false;

    for (int i = 2; i <= number / 2; i++){
        if (number % i == 0)
            return false;
    }
    return true;
}
```

```

nt main()
{
    int number;
    cout << "Enter a number: ";
    cin >> number;
    if (isPrime(number))
        cout << number << " is a Prime number." << endl;
    else
        cout << number << " is not a Prime number." << endl;
    return 0;
}
#include <iostream>
using namespace std;
int main()
{
    int arr[10];
    cout << "Enter 10 integers:" << endl;
    for (int i = 0; i < 10; i++){
        cin >> arr[i];
    }
    cout << "Array elements are:" << endl;
    for (int i = 0; i < 10; i++){
        cout << arr[i] << " ";
    }
    return 0;
}

```

Task 5: Arrays

a) Read 10 integers into an array and display them.

solution:

```
#include <iostream>
using namespace std;
int main()
{
    int arr[10];
    int largest, smallest;
    cout << "Enter 10 integers:" << endl;
    for (int i = 0; i < 10; i++)
    {
        cin >> arr[i];
    }
    largest = smallest = arr[0];
    for (int i = 1; i < 10; i++)
    {
        if (arr[i] > largest)
            largest = arr[i];
        if (arr[i] < smallest)
            smallest = arr[i];
    }
    cout << "Largest element = " << largest << endl;
    cout << "Smallest element = " << smallest << endl;
    return 0;
}
```

b) Find largest and smallest elements.

solution

```
#include <iostream>
using namespace std;
int main()
{int arr[10];
int largest, smallest;
cout << "Enter 10 integers:" << endl;
for (int i = 0; i < 10; i++)
{
    cin >> arr[i];
}
largest = smallest = arr[0];
for (int i = 1; i < 10; i++)
{
    if (arr[i] > largest)
        largest = arr[i];
    if (arr[i] < smallest)
        smallest = arr[i];
}
cout << "Largest element = " << largest << endl;
cout << "Smallest element = " << smallest << endl;
return 0;
}
```

c) Calculate sum and average.

solution

```
#include <iostream>
using namespace std;
int main()
{
int arr[10];
int sum = 0;
float average;
cout << "Enter 10 integers:" << endl;
for (int i = 0; i < 10; i++)
{
    cin >> arr[i];
    sum = sum + arr[i];
}
average = sum / 10.0;
cout << "Sum = " << sum << endl;
cout << "Average = " << average << endl;
return 0;
}
```

d) Search an element using Linear Search.
solution

```
#include <iostream>
using namespace std;
int main()
{
    int arr[10];
    int searchElement;
    bool found = false;
    cout << "Enter 10 integers:" << endl;
    for (int i = 0; i < 10; i++) {
        cin >> arr[i];
    }
    cout << "Enter element to search: ";
    cin >> searchElement;
    for (int i = 0; i < 10; i++){
        if (arr[i] == searchElement){
            cout << "Element found at position " << i + 1 << endl;
            found = true;
            break;
        }
    }
    if (!found){
        cout << "Element not found." << endl;
    }
    return 0;
}
```